

T Deekshith

192472253

CSA6501-GENERATIVE AI

LAB PROGRAMS(26-32)

26. General Prompt → AI Explanation

Program

```
from openai import OpenAI

client = OpenAI()

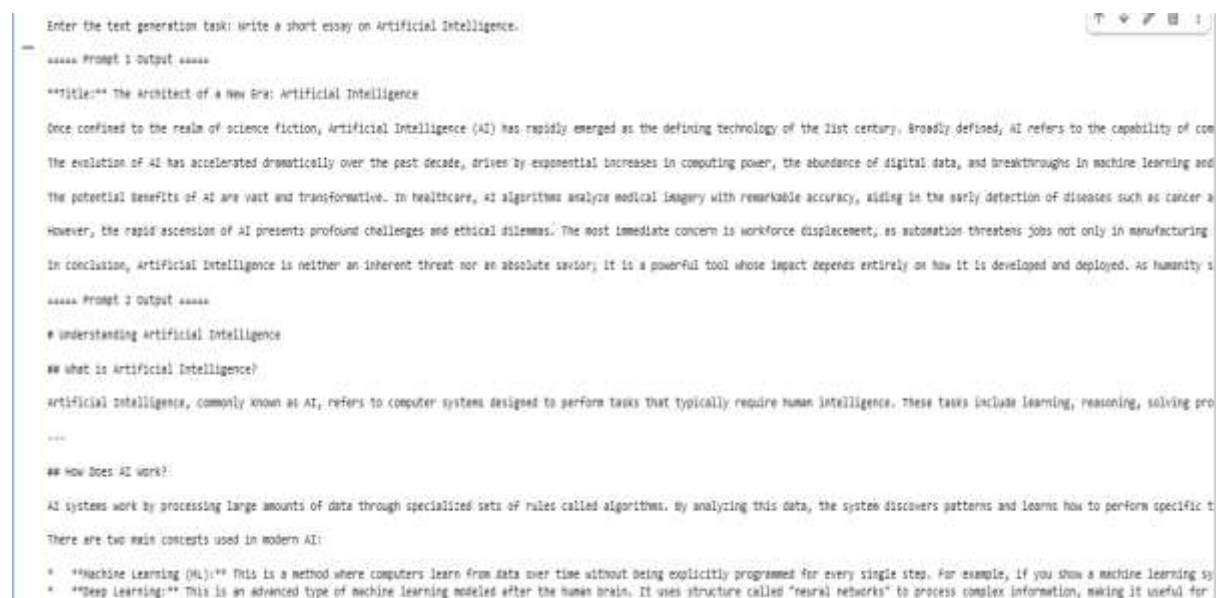
prompt = input("Enter your prompt: ")

response = client.responses.create(
    model="gpt-5.6",
    input=prompt
)

print("\nGenerated Response:\n")

print(response.output_text)
```

Output



```
Enter the text generation task: Write a short essay on Artificial Intelligence.

===== Prompt 1 Output =====

**Title:** The Architect of a New Era: Artificial Intelligence

Once confined to the realm of science fiction, Artificial Intelligence (AI) has rapidly emerged as the defining technology of the 21st century. Broadly defined, AI refers to the capability of com

The evolution of AI has accelerated dramatically over the past decade, driven by exponential increases in computing power, the abundance of digital data, and breakthroughs in machine learning and

the potential benefits of AI are vast and transformative. In healthcare, AI algorithms analyze medical imagery with remarkable accuracy, aiding in the early detection of diseases such as cancer a

However, the rapid ascension of AI presents profound challenges and ethical dilemmas. The most immediate concern is workforce displacement, as automation threatens jobs not only in manufacturing

In conclusion, Artificial Intelligence is neither an inherent threat nor an absolute savior; it is a powerful tool whose impact depends entirely on how it is developed and deployed. As humanity s

===== Prompt 2 Output =====

# Understanding Artificial Intelligence

## What is Artificial Intelligence?

Artificial Intelligence, commonly known as AI, refers to computer systems designed to perform tasks that typically require human intelligence. These tasks include learning, reasoning, solving pro

...

## How Does AI Work?

AI systems work by processing large amounts of data through specialized sets of rules called algorithms. By analyzing this data, the system discovers patterns and learns how to perform specific t

There are two main concepts used in modern AI:

* **Machine Learning (ML):** This is a method where computers learn from data over time without being explicitly programmed for every single step. For example, if you show a machine learning sy
* **Deep Learning:** This is an advanced type of machine learning modeled after the human brain. It uses structure called "neural networks" to process complex information, making it useful for
```

27. Topic-Based Text Generation

```
from openai import OpenAI

client = OpenAI()

topic = input("Enter a topic: ")

prompt = f"""

Write a clear and detailed explanation about the following topic:

{topic}

"""

response = client.responses.create(
    model="gpt-5.6",
    input=prompt
)

print("\nGenerated Response:\n")
print(response.output_text)
```

Output



```
*** Enter a topic: Importance of Python Programming
Python has emerged as a cornerstone of modern technology, largely due to its unparalleled versatility, readability, and powerful ecosystem. Featuring a clean, English-like
```

28. Explain a Topic in Simple Words

```
from openai import OpenAI

client = OpenAI()

user_prompt = input("Enter your prompt: ")

prompt = f"""

Explain the following topic in simple and easy words.

Use examples wherever necessary.

Topic:

{user_prompt}

"""

response = client.responses.create(
    model="gpt-5.6",
```

```

    input=prompt
)
print("\nGenerated Response:\n")
print(response.output_text)

```

Output



```

Enter the Python problem: write a python program to check whether a number is prime.
Generated Python Code:
'''python
import math

def is_prime(number):
    """checks if a given integer is a prime number.

    Parameters:
        number (int): The number to check.

    Returns:
        bool: True if the number is prime, False otherwise.
    """
    # Prime numbers are greater than 1
    if number <= 1:
        return False

    # 2 is the only even prime number
    if number == 2:
        return True

    # exclude all other even numbers
    if number % 2 == 0:
        return False

    # Check odd factors up to the square root of the number
    square_root = int(math.sqrt(number))
    for i in range(3, square_root + 1, 2):
        if number % i == 0:
            return False
    return True
'''

```

29. Python Code Generation

```

from openai import OpenAI

client = OpenAI()

problem = input("Enter the Python problem: ")

prompt = f"""

Generate a correct Python program for the following problem.

Problem:

{problem}

Provide only the Python code with proper comments.

"""

response = client.responses.create(
    model="gpt-5.6",
    input=prompt
)

print("\nGenerated Python Code:\n")
print(response.output_text)

```

Output

```
*** Enter Database Schema: Student(StudentID, Name, Department, CGPA)
Enter Requirement: Display students with CGPA greater than 8.5.
```

Generated SQL Query:

```
```sql
SELECT * FROM Student WHERE CGPA > 8.5;
```
```

30. SQL Query Generation

```
from openai import OpenAI
```

```
client = OpenAI()
```

```
schema = input("Enter Database Schema: ")
```

```
request = input("Enter your request: ")
```

```
prompt = f
```

```
response = client.responses.create(
```

```
    model="gpt-5.6",
```

```
    input=prompt
```

```
)
```

```
print("\nGenerated SQL Query:\n")
```

```
print(response.output_text)
```

Output

```
*** Enter your prompt: Explain Machine Learning in simple words.

Generated Response:

Imagine you want to teach a child how to recognize a "dog".

You wouldn't sit them down and explain: "A dog is a quadrupedal mammal with a snout, non-retractable claws, and a tail." That's too complicated!

Instead, you just point at a dog and say, "Look, a dog!" After seeing enough dogs-big ones, small ones, fluffy ones-the child's brain figures out the pattern. Eventually

"Machine Learning is doing the exact same thing, but with a computer."

***

*** Traditional Programming vs. Machine Learning

* "Traditional Programming:" A human writes precise, step-by-step instructions for the computer to follow.
  * "Formula:" Data + Rules = Answer.
* "Machine Learning:" You give the computer a huge pile of data and the correct answers, and the computer "figures out the rules on its own".
  * "Formula:" Data + Answers = Rules.

*** How it works in 3 simple steps:

1. "Feed it Data (The Examples):" You show the computer thousands of photos of cats and dogs.
2. "Look for Patterns (The Practice):" The computer analyzes the photos and notices things like, "Cats usually have pointy ears" or "Dogs usually have longer snouts."
3. "Make Predictions (The Test):" You show it a brand-new photo. Based on the patterns it learned, the computer guesses, "That's a cat!" If it gets it wrong, it adjust
```

31. Zero-Shot, One-Shot and Few-Shot Prompting

PROGRAM

```
client = OpenAI()

task = input("Enter task: ")

# Zero-shot
zero_shot = f"""
{task}
"""

response1 = client.responses.create(
    model="gpt-5.6",
    input=zero_shot
)

# One-shot
one_shot = f"""
Translate English words into French.
English: Hello
French: Bonjour
Now perform this task:
{task}
"""

response2 = client.responses.create(
    model="gpt-5.6",
    input=one_shot
)

# Few-shot
few_shot = f"""
Translate English words into French.
English: Hello
French: Bonjour
English: Thank You
French: Merci
"""
```

English: Good Night

French: Bonne nuit

Now perform this task:

{task}

"""

```
response3 = client.responses.create(
```

```
    model="gpt-5.6",
```

```
    input=few_shot
```

```
)
```

```
print("\n----- Zero-shot -----")
```

```
print(response1.output_text)
```

```
print("\n----- One-shot -----")
```

```
print(response2.output_text)
```

```
print("\n----- Few-shot -----")
```

```
print(response3.output_text)
```

Output

```
... Enter task: Translate 'Good Morning' into French.
```

```
----- Zero-shot -----
```

```
"Good morning" in French is **Bonjour**.
```

```
----- One-shot -----
```

```
French: Bonjour
```

```
----- Few-shot -----
```

```
French: Bonjour
```

32. Comparing Two Different Prompts

```
from openai import OpenAI

client = OpenAI()

task = input("Enter the text generation task: ")

# Prompt 1
prompt1 = f"""
{task}

Write the response as a well-structured essay with:

- Title

- Introduction

- Main content

- Conclusion

"""

response1 = client.responses.create(
    model="gpt-5.6",
    input=prompt1
)

# Prompt 2
prompt2 = f"""
{task}

Explain the topic in a structured educational format using:

1. Introduction

2. What is the topic?

3. How does it work?

4. Main concepts

5. Applications

6. Advantages

7. Conclusion

Use simple language and clear headings.

"""

response2 = client.responses.create(
```

```

model="gpt-5.6",

input=prompt2
)

print("\n===== Prompt 1 Output =====\n")

print(response1.output_text)

print("\n===== Prompt 2 Output =====\n")

print(response2.output_text)

```

Output



```

Enter your prompt: what is Artificial Intelligence?
Generated Response:

At its most basic level, "Artificial Intelligence (AI)" refers to the development of computer systems capable of performing tasks that historically required human intelligence. These tasks include things like "reasoning, learning from experience, solving complex problems, understanding language, recognizing patterns, and making decisions."

Instead of just following a rigid, step-by-step program written by a human, AI systems process vast amounts of data, find patterns within that data, and make predictions or decisions based on those patterns.

How AI Works: The Core Concepts
To understand AI, it helps to know how the technology is layered:

1. "Artificial Intelligence (The Umbrella Term)": Any technique that enables computers to mimic human intelligence.
2. "Machine Learning (ML)": A subset of AI where computers are fed massive amounts of data and "learn" how to perform a task on their own, improving over time without being explicitly programmed.
3. "Deep Learning": A specialized, advanced form of Machine Learning that uses "neural networks"—math models loosely inspired by the structure of the human brain—to process data.
4. "Generative AI": A cutting-edge type of Deep Learning capable of "creating" new content—such as text (e.g., ChatGPT), images (e.g., MidJourney), music, or code—based on the patterns it has learned.

The Stages of AI Capability
Experts generally divide AI into three categories based on its capabilities:

* "Narrow AI (Weak AI)": AI designed to perform a specific, single task extremely well. "All AI that exists today is Narrow AI." Examples include chess-playing computers, recommendation algorithms, and voice assistants.
* "Artificial General Intelligence (AGI / Strong AI)": A theoretical form of AI that would possess human-level intelligence across "all" domains. An AGI could learn, adapt, and solve any problem that a human can.
* "Artificial Superintelligence (ASI)": A theoretical future AI that far surpasses human intelligence in every way—creative, social, scientific, and strategic.

```